Self Learnign Assignment Day 2
C++ Programming

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6. Create Address class with data members
HouseNo
Colony
Area
City
Pincode
Implement followings
- **Default constructor**
- **Parameterized constructor**
- **Accept Function for data input**
- **Display function for data display**
- **Separate header and implementation files**
- **Write function to find whether two addresses are same or different.**
- **Overlaod == operator**

Program:
address.h

```
#include <iostream>
using namespace std;

class Address
{
private:
   int houseNo;
   string colony;
   string area;
   string city;
   int pincode;

public:
   // default Constructor
   Address();

   // parameterized Constructor
   Address(int, string, string, string, int);
```

```
   // accept Function
   void accept();

   // display Function
   void display() const;

   // function to compare addresses
   bool isSame(Address);

   // overloaded == operator
   bool operator==(Address);
};
```

address.cpp

```
#include<iostream>
#include<cstring>
#include "address.h"
using namespace std;

// default Constructor
Address::Address()
{
   houseNo = 0;
   colony = "";
   area = "";
   city = "";
   pincode = 0;
}

// parameterized Constructor
Address::Address(int h, string c, string a, string ci, int p)
{
   houseNo = h;
   colony = c;
   area = a;
   city = ci;
   pincode = p;
}

// accept Function
void Address::accept()
{
   cout << "Enter House Number: ";
   cin >> houseNo;

   cout << "Enter Colony: ";
```

```
    cin >> colony;

    cout << "Enter Area: ";
    cin >> area;

    cout << "Enter City: ";
    cin >> city;

    cout << "Enter Pincode: ";
    cin >> pincode;
}

// display Function
void Address::display() const
{
    cout << "House No: " << houseNo << endl;
    cout << "Colony: " << colony << endl;
    cout << "Area: " << area << endl;
    cout << "City: " << city << endl;
    cout << "Pincode: " << pincode << endl;
}

// function to check if addresses are same
bool Address::isSame(Address a)
{
    if (houseNo == a.houseNo &&
        colony == a.colony &&
        area == a.area &&
        city == a.city &&
        pincode == a.pincode)
        return true;
    else
        return false;
}

// operator Overloading ==
bool Address::operator==(Address a)
{
    return (houseNo == a.houseNo &&
            colony == a.colony &&
            area == a.area &&
            city == a.city &&
            pincode == a.pincode);
}
```

testAddress.cpp

```
#include "address.h"
using namespace std;

int main()
{
   Address a1, a2;

   cout << "Enter First Address:\n";
   a1.accept();

   cout << "\nEnter Second Address:\n";
   a2.accept();

   cout << "\nFirst Address:\n";
   a1.display();

   cout << "\nSecond Address:\n";
   a2.display();

   // Using function
   if(a1.isSame(a2))
       cout << "\nAddresses are SAME\n";
   else
       cout << "\nAddresses are DIFFERENT\n";

   // Using overloaded operator
   if(a1 == a2)
       cout << "\n(Using ==) Addresses are SAME\n";
   else
       cout << "\n(Using ==) Addresses are DIFFERENT\n";

   return 0;
}
```

Compiling command:
$ g++ address.cpp testAddress.cpp -o address.o

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_Abhin
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_Abhin
Enter First Address:
Enter House Number: 22
Enter Colony: defence colony
Enter Area: Enter City: agra
Enter Pincode: 282001

Enter Second Address:
Enter House Number: 25
Enter Colony: Defence Colony
Enter Area: Enter City: agra
Enter Pincode: 282001

First Address:
House No: 22
Colony: defence
Area: colony
City: agra
Pincode: 282001

Second Address:
House No: 25
Colony: Defence
Area: Colony
City: agra
Pincode: 282001

Addresses are DIFFERENT

(Using ==) Addresses are DIFFERENT
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_Abhin
```

7. Create Payslip class with data members
EmpNo
EmpName
DA
HRA
BasicSalary
Implement followings

- **Default constructor**
- **Parameterized constructor**
- **Accept Function for data input**
- **Display function for data display**
- **Function to calculate salary from DA, HRA and Basic Salary**
- **Separate header and implementation files**

Program:
payslip.h

```
#include <iostream>
using namespace std;

class Payslip
{
private:
   int empNo;
   string empName;
   float DA;
   float HRA;
   float basicSalary;

public:
   // default Constructor
   Payslip();

   // parameterized Constructor
   Payslip(int, string, float, float, float);

   // accept Function
   void accept();

   // display Function
   void display();

   // function to calculate total salary
   float calculateSalary();
};
```

payslip.cpp

```
#include "payslip.h"

// default Constructor
Payslip::Payslip()
{
```

```
   empNo = 0;
   empName = "";
   DA = 0;
   HRA = 0;
   basicSalary = 0;
}

// parameterized Constructor
Payslip::Payslip(int no, string name, float da, float hra, float basic)
{
   empNo = no;
   empName = name;
   DA = da;
   HRA = hra;
   basicSalary = basic;
}

// accept Function
void Payslip::accept()
{
   cout << "Enter Employee No: ";
   cin >> empNo;

   cout << "Enter Employee Name: ";
   cin >> empName;

   cout << "Enter Basic Salary: ";
   cin >> basicSalary;

   cout << "Enter DA: ";
   cin >> DA;

   cout << "Enter HRA: ";
   cin >> HRA;
}

// calculate Salary
float Payslip::calculateSalary()
{
   return basicSalary + DA + HRA;
}

// display Function
void Payslip::display()
{
```

```
    cout << "Employee No: " << empNo << endl;
    cout << "Employee Name: " << empName << endl;
    cout << "Basic Salary: " << basicSalary << endl;
    cout << "DA: " << DA << endl;
    cout << "HRA: " << HRA << endl;
    cout << "Total Salary: " << calculateSalary() << endl;
}
```

testpayslip.cpp

```
#include "payslip.h"

int main()
{
    Payslip p1;

    cout << "Enter Employee Details\n";
    p1.accept();

    cout << "\nEmployee Payslip\n";
    p1.display();

    return 0;
}
```

Command to compile:
$ g++ payslip.cpp testpayslip.cpp -o payslip.o

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) % g++ payslip.cp
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) % ./payslip.o
Enter Employee Details
Enter Employee No: 101
Enter Employee Name: ankit
Enter Basic Salary: 20000
Enter DA: 2900
Enter HRA: 5000

Employee Payslip
Employee No: 101
Employee Name: ankit
Basic Salary: 20000
DA: 2900
HRA: 5000
Total Salary: 27900
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_AbhinayKumar(260240120005) %
```

8. Create MobilePhone class with data members
ModelNo
Price
Manufacturer
Quantity

Implement followings
• Default constructor
• Parameterized constructor
• Accept Function for data input
• Display function for data display
• Function to search MobilePhone model availability
• Write menu driven program with following menus
o Add Mobile
o Display List of available mobiles
o Find out quantity of particular mobile type
o Find out availability of Particular model

Program:

mobilephone.h

```
#include <iostream>
using namespace std;

class MobilePhone
{
private:
   string modelNo;
   float price;
   string manufacturer;
   int quantity;

public:
   // default constructor
   MobilePhone();

   // parameterized constructor
   MobilePhone(string, float, string, int);

   // accept data
   void accept();

   // display data
   void display();

   // search model availability
   bool searchModel(string);

   // get functions
   string getModel();
   int getQuantity();
```

```
};
```

mobilephone.cpp

```
#include "mobilephone.h"

// default constructor
MobilePhone::MobilePhone()
{
   modelNo = "";
   price = 0;
   manufacturer = "";
   quantity = 0;
}

// parameterized constructor
MobilePhone::MobilePhone(string m, float p, string manu, int q)
{
   modelNo = m;
   price = p;
   manufacturer = manu;
   quantity = q;
}

// accept function
void MobilePhone::accept()
{
   cout << "Enter Model No: ";
   cin >> modelNo;

   cout << "Enter Price: ";
   cin >> price;

   cout << "Enter Manufacturer: ";
   cin >> manufacturer;

   cout << "Enter Quantity: ";
   cin >> quantity;
}

// display function
void MobilePhone::display()
{
   cout << "Model No: " << modelNo << endl;
   cout << "Price: " << price << endl;
```

```
    cout << "Manufacturer: " << manufacturer << endl;
    cout << "Quantity: " << quantity << endl;
}

// search model
bool MobilePhone::searchModel(string m)
{
    return modelNo == m;
}

string MobilePhone::getModel()
{
    return modelNo;
}

int MobilePhone::getQuantity()
{
    return quantity;
}
```

testmobilephone.cpp

```
#include "mobilephone.h"

int main()
{
    MobilePhone m[100];
    int count = 0;
    int choice;
    string model;

    do
    {
        cout << "\n--- Mobile Store Menu ---\n";
        cout << "1. Add Mobile\n";
        cout << "2. Display Mobiles\n";
        cout << "3. Find Quantity of Particular Mobile\n";
        cout << "4. Check Model Availability\n";
        cout << "5. Exit\n";
        cout << "Enter choice: ";
        cin >> choice;

        switch(choice)
        {
            case 1:
                m[count].accept();
                count++;
```

```
            break;

        case 2:
            for(int i=0;i<count;i++)
                m[i].display();
            break;

        case 3:
            cout << "Enter Model No: ";
            cin >> model;
            for(int i=0;i<count;i++)
            {
                if(m[i].getModel()==model)
                    cout << "Quantity: " << m[i].getQuantity() << endl;
            }
            break;

        case 4:
            cout << "Enter Model No: ";
            cin >> model;

            for(int i=0;i<count;i++)
            {
                if(m[i].searchModel(model))
                {
                    cout << "Model Available\n";
                    break;
                }
            }
            break;
    }

  }while(choice!=5);

  return 0;
}
```

Command to compile:

$ g++ mobilephone.cpp testmobilephone.cpp -o mobilephone.o

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_

--- Mobile Store Menu ---
1. Add Mobile
2. Display Mobiles
3. Find Quantity of Particular Mobile
4. Check Model Availability
5. Exit
Enter choice: 1
Enter Model No: iphone 16
Enter Price: Enter Manufacturer: apple
Enter Quantity: 100

--- Mobile Store Menu ---
1. Add Mobile
2. Display Mobiles
3. Find Quantity of Particular Mobile
4. Check Model Availability
5. Exit
Enter choice: 3
Enter Model No: iphone 16
Quantity: 100

--- Mobile Store Menu ---
1. Add Mobile
2. Display Mobiles
3. Find Quantity of Particular Mobile
4. Check Model Availability
5. Exit
Enter choice:
--- Mobile Store Menu ---
1. Add Mobile
2. Display Mobiles
3. Find Quantity of Particular Mobile
4. Check Model Availability
```

```
(base) abhinyaay@Abhinays-MacBook-Air SLA_Day1_
--- Mobile Store Menu ---
1. Add Mobile
2. Display Mobiles
3. Find Quantity of Particular Mobile
4. Check Model Availability
5. Exit
Enter choice: 4
Enter Model No: iphone 16
Model Available

--- Mobile Store Menu ---
1. Add Mobile
2. Display Mobiles
3. Find Quantity of Particular Mobile
4. Check Model Availability
5. Exit
Enter choice:
--- Mobile Store Menu ---
1. Add Mobile
2. Display Mobiles
3. Find Quantity of Particular Mobile
4. Check Model Availability
5. Exit
Enter choice: 2
Model No: iphone
Price: 16
Manufacturer: apple
Quantity: 100

--- Mobile Store Menu ---
1. Add Mobile
2. Display Mobiles
3. Find Quantity of Particular Mobile
4. Check Model Availability
5. Exit
Enter choice:
```